Power BI Task: WEEK 9

Housing Data Cleaning & Transformation

-done by **Sai Meera R** from **B.sc Data Science & AI**

Introduction

The objective of this assignment is to clean and transform the King County House Sales Dataset using Power BI. The dataset contains historical house sales information, including property prices, bedrooms, bathrooms, living area, construction year, and renovation details.

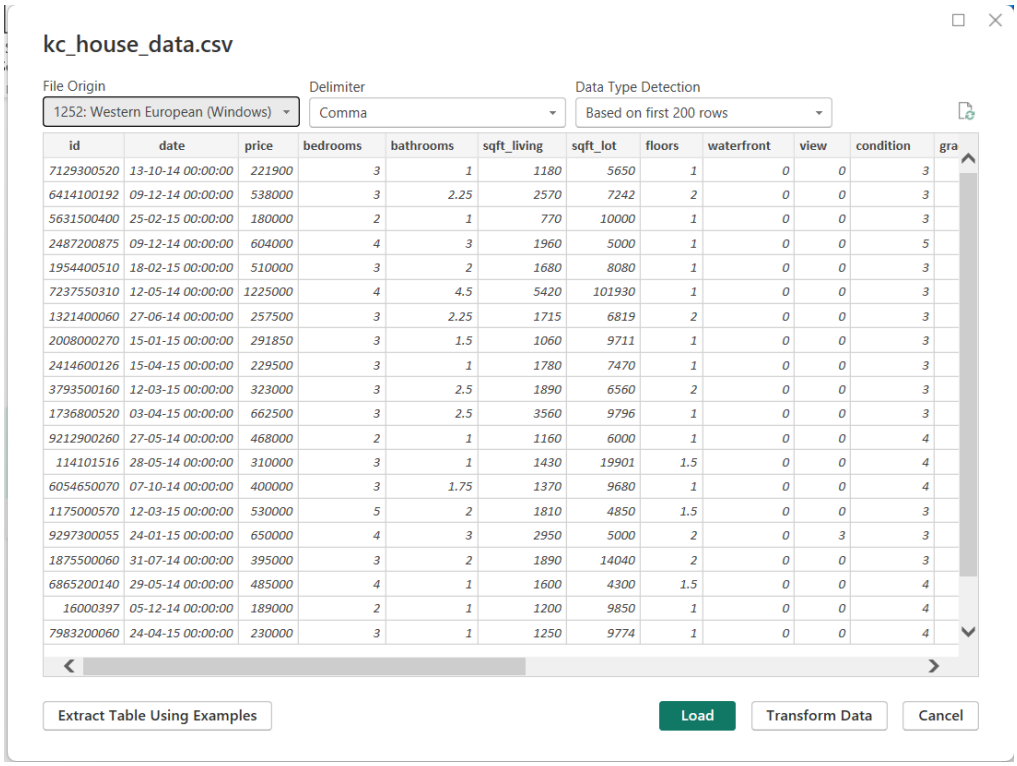
Power Query Editor was used to clean the dataset, correct data types, create calculated columns, and prepare the data for further analysis.

Data Import and Transformation

The CSV dataset was imported into Power BI using the Get Data option. The Transform Data option was selected to open Power Query Editor.

The data types of important columns were checked and corrected. The sale date column was converted from Text to Date/Time using the English (United Kingdom) locale to ensure correct date interpretation.

Imported dataset in Power Query Editor



id	date	price	bedrooms	bathrooms	sqft_living	sqft_lot	floors	waterfront	view	condition	gra
7129300520	13-10-14 00:00:00	221900	3	1	1180	5650	1	0	0	3	
6414100192	09-12-14 00:00:00	538000	3	2.25	2570	7242	2	0	0	3	
5631500400	25-02-15 00:00:00	180000	2	1	770	10000	1	0	0	3	
2487200875	09-12-14 00:00:00	604000	4	3	1960	5000	1	0	0	5	
1954400510	18-02-15 00:00:00	510000	3	2	1680	8080	1	0	0	3	
7237550310	12-05-14 00:00:00	1225000	4	4.5	5420	101930	1	0	0	3	
1321400060	27-06-14 00:00:00	257500	3	2.25	1715	6819	2	0	0	3	
2008000270	15-01-15 00:00:00	291850	3	1.5	1060	9711	1	0	0	3	
2414600126	15-04-15 00:00:00	229500	3	1	1780	7470	1	0	0	3	
3793500160	12-03-15 00:00:00	323000	3	2.5	1890	6560	2	0	0	3	
1736800520	03-04-15 00:00:00	662500	3	2.5	3560	9796	1	0	0	3	
9212900260	27-05-14 00:00:00	468000	2	1	1160	6000	1	0	0	4	
114101516	28-05-14 00:00:00	310000	3	1	1430	19901	1.5	0	0	4	
6054650070	07-10-14 00:00:00	400000	3	1.75	1370	9680	1	0	0	4	
1175000570	12-03-15 00:00:00	530000	5	2	1810	4850	1.5	0	0	3	
9297300055	24-01-15 00:00:00	650000	4	3	2950	5000	2	0	3	3	
1875500060	31-07-14 00:00:00	395000	3	2	1890	14040	2	0	0	3	
6865200140	29-05-14 00:00:00	485000	4	1	1600	4300	1.5	0	0	4	
16000397	05-12-14 00:00:00	189000	2	1	1200	9850	1	0	0	4	
7983200060	24-04-15 00:00:00	230000	3	1	1250	9774	1	0	0	4	

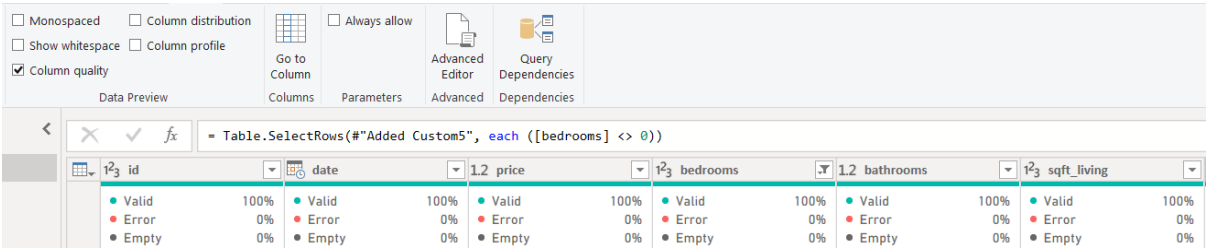
Data Cleaning

The following data-cleaning operations were performed:

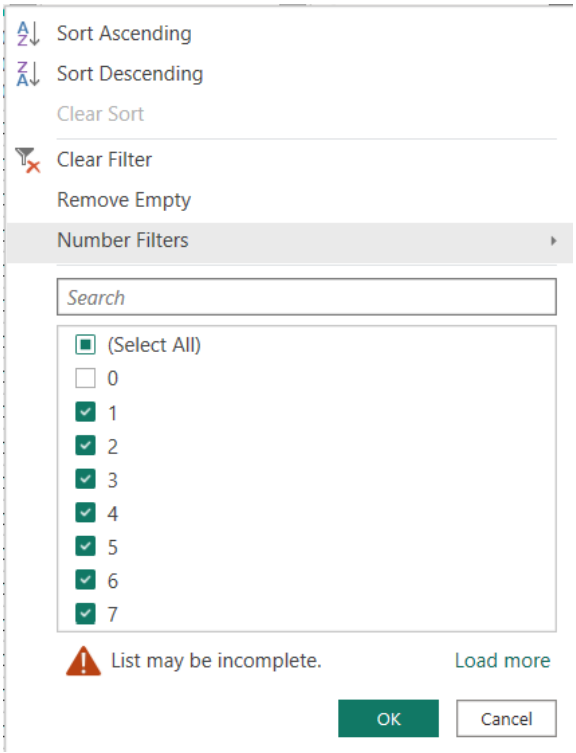
- Checked column quality using Column Quality indicators.
- Verified valid, error, and empty values.
- Corrected the data types of numerical and date columns.
- Filtered out records with zero bedrooms.
- Filtered out records with zero bathrooms.
- Removed null values from the bedroom and bathroom columns where present.

Valid fractional bathroom values such as 0.75, 1.5, and 2.25 were retained.

Column Quality indicators



Bedroom and bathroom filtering



Creation of Derived Columns

Custom columns were created in Power Query Editor to extract useful information from the existing dataset.

- **Sale Year**

Extracted the year from the sale date to identify the year in which each property was sold.

- **Sale Month**

Extracted the month from the sale date to support monthly sales analysis.

- **Property Age at Sale**

Calculated the age of the property at the time of sale.

Formula:

[Sale Year] - [yr_built]

- **Price per SqFt**

Calculated the price per square foot using the house price and living area.

Formula:

[price] / [sqft_living]

- **Total Area**

Calculated the total area by combining living area and basement area.

Formula:

[sqft_living] + [sqft_basement]

- **Is Renovated**

Created a categorical column to identify whether a renovation year was recorded.

Formula:

if [yr_renovated] = 0 then "No" else "Yes"

Custom columns in Power Query Editor

Sale year	Sale Month	Property Age at Sale	Price per Sqft
Valid 100%	Valid 100%	Valid 100%	Valid 100%
Error 0%	Error 0%	Error 0%	Error 0%
Empty 0%	Empty 0%	Empty 0%	Empty 0%
2014	10	59	188.0508475
2014	12	63	209.3385214
2015	2	82	233.7662338
2014	12	49	308.1632653
2015	2	28	303.5714286
2014	5	13	226.0147601
2014	6	19	150.1457726
2015	1	52	275.3301887
2015	4	55	128.9325843
2015	3	12	170.8994709
2015	4	50	186.0955056
2014	5	72	403.4482759
2014	5	87	216.7832168
2014	10	37	291.9708029
2015	3	115	292.8176796
2015	1	36	220.3389831
2014	7	20	208.994709
2014	5	98	303.125
2014	12	93	157.5

Total Area	Is Renovated
Valid 100%	Valid 100%
Error 0%	Error 0%
Empty 0%	Empty 0%
1180	No
2970	Yes
770	No
2870	No
1680	No
6950	No
1715	No
1060	No
2510	No
1890	No
5260	No
1460	No
1430	No
1370	No
1810	No
3920	No
1890	No
1600	No
1200	No

Property Categorization

Properties were categorized based on their total area.

Total Area (sq. ft.)	Category
Below 1,000	Small
1,000–1,999	Medium
2,000–3,999	Large
4,000 and above	Luxury

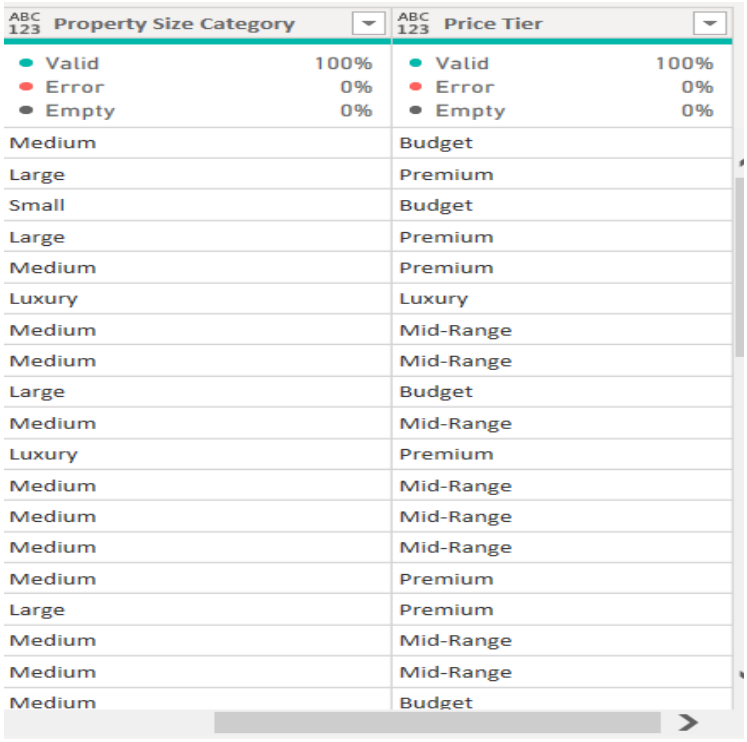
A new column named Property Size Category was created using conditional logic.

House prices were also categorized into four price tiers.

House Price	Price Tier
Below $250,000	Budget
$250,000–$499,999	Mid-Range
$500,000–$999,999	Premium
$1,000,000 and above	Luxury

A new column named Price Tier was created to classify properties based on their prices.

Property Size Category and Price Tier columns



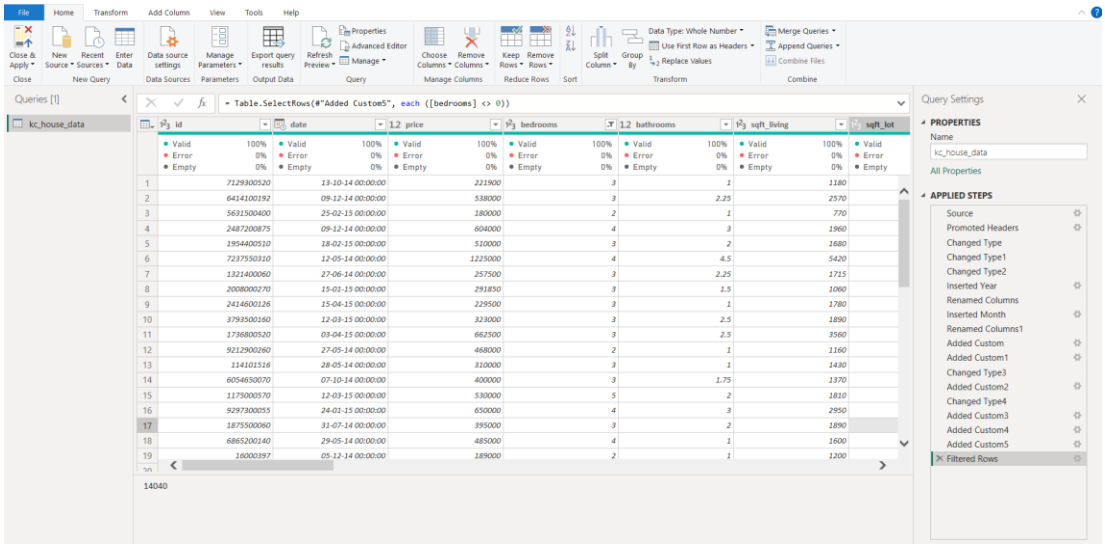
Property Size Category	Price Tier
Valid 100%	Valid 100%
Error 0%	Error 0%
Empty 0%	Empty 0%
Medium	Budget
Large	Premium
Small	Budget
Large	Premium
Medium	Premium
Luxury	Luxury
Medium	Mid-Range
Medium	Mid-Range
Large	Budget
Medium	Mid-Range
Luxury	Premium
Medium	Mid-Range
Medium	Mid-Range
Medium	Mid-Range
Medium	Premium
Large	Premium
Medium	Mid-Range
Medium	Mid-Range
Medium	Budget

Loading the Transformed Data

After completing the cleaning and transformation operations, the Close & Apply option was selected to load the prepared dataset into Power BI.

The transformed housing data was successfully loaded into the main Power BI Desktop window.

Final loaded dataset and Fields pane



Conclusion

In this assignment, the King County House Sales Dataset was successfully imported, cleaned, and transformed using Power BI.

New columns were created to identify sale years, sale months, property ages, price per square foot, total area, and renovation status. Properties were also classified into different size categories and price tiers.

These transformations prepare the dataset for further real-estate analysis and dashboard development in the next assignment.